

- A **file descriptor (FD)** is simply an integer number that the operating system gives to a process when it opens something.
- The operating system assigns the lowest available file descriptor number when you open a resource.

Think of it as a ticket number.

Suppose you tell Linux:

Open **student.txt**

Linux may say:

Okay, I will give you ticket number 3.

So:

Student.txt —> **FD3**

Your program doesn't have to keep saying "**student.txt**" every time.

It can simply say FD 3 to refer to that open file.

Similarly, Linux gives your program a number:

File → File Descriptor

For example: student.txt → 3

That number 3 is the file descriptor.

Every normal process starts with three standard file descriptors:

0 → Standard Input

1 → Standard Output

2 → Standard Error

0 → stdin → Where do I GET input?

- 1 → stdout → Where do I SEND normal output?
2 → stderr → Where do I SEND error output?

We normally write them as:

STDIN_FILENO
STDOUT_FILENO
STDERR_FILENO

Number	Name	Meaning
0	STDIN_FILENO	Input
1	STDOUT_FILENO	Output
2	STDERR_FILENO	Error output

Program 1: stdout — Standard Output

```
#include <unistd.h>

int main()
{
    write(STDOUT_FILENO, "Hello World\n", 12);
    return 0;
}
```

```
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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
```

Program 2: stdin — Standard Input

```
#include <unistd.h>
```

```

int main()
{
    char buffer[20];
    read(STDIN_FILENO, buffer, sizeof(buffer));
    write(STDOUT_FILENO, buffer, sizeof(buffer));
    return 0;
}

```

```

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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi

```

Program 3: Proper stdin + stdout

```

#include <stdio.h>
#include <unistd.h>

int main()
{
    char buffer[100];
    int n;
    printf("Enter something: ");
    n = read(STDIN_FILENO, buffer, sizeof(buffer) - 1);
    if (n > 0)
    {
        buffer[n] = '\0';
        write(STDOUT_FILENO, buffer, n);
    }
    return 0;
}

```

```

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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program3.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program3.c -o program3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program3
hi dharshan
hi dharshan
Enter something: dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi

```

Program 4: stderr — Standard Error

```

#include <unistd.h>

int main()
{
    write(STDERR_FILENO, "This is an error\n", 17);
    return 0;
}

```

As `STDERR_FILENO` is: 2

`write(STDERR_FILENO, "This is an error\n", 17)` is equivalent to:

```
write(2, "This is an error\n", 17);
```

You will see output as: This is an error.

```

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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program3.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program3.c -o program3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program3
hi dharshan
hi dharshan
Enter something: dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program4.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program4.c -o program4
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program4
This is an error
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ |

```

Program 5: stderr for an Actual Error

```

#include <stdio.h>

#include <fcntl.h>

#include <unistd.h>

int main()
{
    int fd;
    fd = open("doesnotexist.txt", O_RDONLY);
    if (fd == -1)
    {
        perror("open");
        return 1;
    }
}

```

```
close(fd);
return 0;
}
```

```
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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program3.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program3.c -o program3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program3
hi dharshan
hi dharshan
Enter something: dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program4.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program4.c -o program4
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program4
This is an error
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program5.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program5.c -o program5
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program5
open: No such file or directory
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$
```

Program 6 Creating file using file descriptor

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("student.txt", O_CREAT | O_WRONLY, 0644);
    if (fd == -1)
    {
        perror("open");
    }
}
```

```
        return 1;
    }
    printf("File created successfully!\n");
    printf("File descriptor = %d\n", fd);
    close(fd);
    return 0;
}
```

gcc create.c -o create
./create

Output:

File created successfully! File descriptor = 3

fd = open("student.txt", O_CREAT | O_WRONLY, 0644); describes the information below:

"student.txt" → file name

O_CREAT → create if it doesn't exist

O_WRONLY → open for writing

0644 → file permissions

fd → stores the file descriptor

```

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C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program3.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program3.c -o program3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program3
hi dharshan
hi dharshan
Enter something: dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program4.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program4.c -o program4
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program4
This is an error
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program5.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program5.c -o program5
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program5
open: No such file or directory
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program6.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program6.c -o program6
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program6
File created successfully!
File descriptor = 3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ |

```

Program 7

```

#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd;
    fd = open("student.txt", O_WRONLY | O_TRUNC);
    if (fd == -1)
    {
        perror("open");
    }
}

```

```
    return 1;
}
write(fd, "New Data\n", 9);
close(fd);
return 0;
}
```

If a file already has data, O_TRUNC removes all its existing contents when you open it for writing.

First, suppose student.txt contains:

Hello
Welcome
Operating Systems

After open() with O_TRUNC:
Student.txt contains (empty)

DUP () : duplicate a file descriptor.

dup() creates a new file descriptor that refers to the same underlying open file or I/O resource as the original descriptor.

Suppose:

```
int fd1 = open("student.txt", O_WRONLY);
```

If the OS gives:

```
fd1 = 3
```

then:

FD 3 → student.txt

Now we do:

```
int fd2 = dup(fd1);
```

The OS creates another file descriptor, usually:

```
fd2 = 4
```

Now both FD 3 and FD 4 refer to the same open file.

Program 7 - Understand dup()

```
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
int main()
{
    int fd1, fd2;
    fd1 = open("student.txt", O_CREAT | O_WRONLY, 0644);
    if (fd1 == -1)
    {
        perror("open");
        return 1;
    }
    printf("fd1 = %d\n", fd1);
    fd2 = dup(fd1);
    printf("fd2 = %d\n", fd2);
    close(fd1);
    close(fd2);
    return 0;
}
```

Output: fd1 = 3 fd2 = 4

```
C:\Users\dhars>wsl
dhars@DHARSHAN:/mnt/c/Users/dhars$ mkdir filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars$ cd filedescriptors
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program1.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program1.c -o program1
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program1
Hello World
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ 2
2: command not found
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program2.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program2.c -o program2
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program2
hi
hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program3.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program3.c -o program3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program3
hi dharschan
hi dharschan
Enter something: dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ hi
Command 'hi' not found, but can be installed with:
sudo snap install hi
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program4.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program4.c -o program4
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program4
This is an error
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program5.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program5.c -o program5
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program5
open: No such file or directory
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program6.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program6.c -o program6
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program6
File created successfully!
File descriptor = 3
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ nano program7.c
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ gcc program7.c -o program7
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ ./program7
dhars@DHARSHAN:/mnt/c/Users/dhars/filedescriptors$ cat student.txt
New Data
```